

DBMS - PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

Skill Task: DBMS

Week : 6

1. Database Design

1.1 Review Table

Attribute	Data Type	Constraint
REVIEW_ID	NUMBER(5)	Primary Key
CUSTOMER_ID	NUMBER(5)	Foreign Key
PRODUCT_ID	NUMBER(5)	Foreign Key
REVIEW_TEXT	VARCHAR2(500)	NOT NULL
REVIEW_DATE	DATE	NOT NULL

1.2 RATING TABLE

Attribute	Data Type	Constraint
RATING_ID	NUMBER(5)	Primary Key
CUSTOMER_ID	NUMBER(5)	Foreign Key
PRODUCT_ID	NUMBER(5)	Foreign Key
RATING	NUMBER(1)	NOT NULL
RATING_DATE	DATE	NOT NULL

2. Table Relationship

The SAVANA_REVIEW and SAVANA_RATING tables are related to the SAVANA_CUSTOMER and SAVANA_PRODUCT tables.

- SAVANA_CUSTOMER → SAVANA_REVIEW: One customer can write multiple reviews.
- SAVANA_PRODUCT → SAVANA_REVIEW: One product can receive multiple reviews.
- SAVANA_CUSTOMER → SAVANA_RATING: One customer can provide multiple ratings.
- SAVANA_PRODUCT → SAVANA_RATING: One product can receive multiple ratings.

3. Table Creation

3.1 Create Review table

```
SQL> CREATE TABLE SAVANA_REVIEW (  
2     REVIEW_ID NUMBER(5) PRIMARY KEY,  
3     CUSTOMER_ID NUMBER(5),  
4     PRODUCT_ID NUMBER(5),  
5     REVIEW_TEXT VARCHAR2(500) NOT NULL,  
6     REVIEW_DATE DATE NOT NULL  
7 );
```

Table created.

3.2 Rating Create table

```
SQL> CREATE TABLE SAVANA_RATING (  
2     RATING_ID NUMBER(5) PRIMARY KEY,  
3     CUSTOMER_ID NUMBER(5),  
4     PRODUCT_ID NUMBER(5),  
5     RATING NUMBER(1) NOT NULL,  
6     RATING_DATE DATE NOT NULL  
7 );
```

Table created.

4.Review Data Insrtion

```
SQL> INSERT INTO SAVANA_REVIEW
  2 VALUES
  3 (1, 101, 201, 'Very good quality product',
  4 TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_REVIEW
  2 VALUES
  3 (2, 102, 202, 'Good product and nice quality',
  4 TO_DATE('23-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_REVIEW
  2 VALUES
  3 (3, 103, 203, 'Product is worth the price',
  4 TO_DATE('24-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_REVIEW
  2 VALUES
  3 (4, 101, 204, 'Good product but delivery was late',
  4 TO_DATE('25-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_REVIEW
  2 VALUES
  3 (5, 104, 205, 'Excellent product and quality',
  4 TO_DATE('26-09-2026','DD-MM-YYYY'));

1 row created.
```

5.Rating Data Insrtion

```
SQL> INSERT INTO SAVANA_RATING
  2 VALUES
  3 (501, 101, 201, 5,
  4 TO_DATE('22-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_RATING
  2 VALUES
  3 (502, 102, 202, 4,
  4 TO_DATE('23-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_RATING
  2 VALUES
  3 (503, 103, 203, 5,
  4 TO_DATE('24-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_RATING
  2 VALUES
  3 (504, 101, 204, 3,
  4 TO_DATE('25-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SAVANA_RATING
  2 VALUES
  3 (505, 104, 205, 5,
  4 TO_DATE('26-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL> COMMIT;
```

6.Retrieve Product Review Details

```

SQL> SELECT
  2     REVIEW_ID,
  3     CUSTOMER_ID,
  4     PRODUCT_ID,
  5     REVIEW_TEXT,
  6     REVIEW_DATE
  7 FROM SAVANA_REVIEW;

```

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID	REVIEW_TEXT	REVIEW_DA
1	101	201	Very good quality product	22-SEP-26
2	102	202	Good product and nice quality	23-SEP-26
3	103	203	Product is worth the price	24-SEP-26
4	101	204	Good product but delivery was late	25-SEP-26
5	104	205	Excellent product and quality	26-SEP-26

7. Calculate Average Product Ratings

```
SQL> SELECT
  2     PRODUCT_ID,
  3     AVG(RATING) AS AVERAGE_RATING
  4 FROM SAVANA_RATING
  5 GROUP BY PRODUCT_ID;
```

PRODUCT_ID	AVERAGE_RATING
201	5
202	4
203	5
204	3
205	5

8. Identify Highly Rated Products

```
SQL> SELECT
  2     PRODUCT_ID,
  3     AVG(RATING) AS AVERAGE_RATING
  4 FROM SAVANA_RATING
  5 GROUP BY PRODUCT_ID
  6 HAVING AVG(RATING) >= 4;
```

PRODUCT_ID	AVERAGE_RATING
201	5
202	4
203	5
205	5

9. Product Rating Report

```
SQL> SELECT
  2     P.PRODUCT_ID,
  3     P.PRODUCT_NAME,
  4     AVG(R.RATING) AS AVERAGE_RATING
  5 FROM SAVANA_PRODUCT P
  6 JOIN SAVANA_RATING R
  7 ON P.PRODUCT_ID = R.PRODUCT_ID
  8 GROUP BY
  9     P.PRODUCT_ID,
 10     P.PRODUCT_NAME;
```

```
no rows selected
```

